

Integration - Set 2

TMUA practice questions by topic. Specification MM7.1-MM7.6. 10 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Let $I = \int_0^1 (x^2 + 1) dx$ and $J = \int_0^1 (x + 1)^2 dx$. Which of the following is true?
- A $I < J$ and $J - I = \frac{4}{3}$
- B $I < J$ and $J - I = \frac{3}{2}$
- C $I = J$
- D $I < J$ and $J - I = 1$
- E $I > J$
- 2 Evaluate $\int_{-1}^2 |x| dx$.
- A 2
- B $\frac{7}{2}$
- C 3
- D $\frac{5}{2}$
- E $\frac{3}{2}$
- F $\frac{9}{2}$
- 3 The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} dx$. Which of the following is true for every n ?
- A The estimate is an overestimate.
- B The estimate is an underestimate.
- C The estimate is an underestimate only when n is even.
- D The estimate is exact.
- E It cannot be decided without calculating.

- 4 Evaluate $\int_1^{43x^2-5} \frac{1}{\sqrt{x}} dx$.
- A $\frac{146}{5}$
 B $\frac{186}{5}$
 C $\frac{28}{5}$
 D $\frac{86}{5}$
 E $\frac{126}{5}$
 F $\frac{136}{5}$
- 5 Find the area of the finite region enclosed by the curve $y = x^2$ and the line $y = 2x + 3$.
- A $\frac{16}{3}$
 B $\frac{28}{3}$
 C $\frac{12}{3}$
 D $\frac{20}{3}$
 E $\frac{32}{3}$
 F $\frac{64}{3}$
- 6 The region bounded by $y = 4 - x^2$, the x -axis and the lines $x = 0$ and $x = 3$ is shaded. Find the total shaded area.
- A $\frac{3}{7}$
 B $\frac{16}{3}$
 C $\frac{25}{3}$
 D $\frac{23}{3}$
 E $\frac{3}{9}$
 F $\frac{9}{9}$
- 7 The trapezium rule is used to estimate $\int_a^b f(x) dx$. For which of the following functions is the estimate guaranteed to be an underestimate for every choice of $a < b$?
- A $f(x) = -x^2$
 B $f(x) = x^4$
 C $f(x) = 2x + 1$
 D $f(x) = x^2$
 E $f(x) = x^3$

- 8** The area between the curve $y = x^2$ and the line $y = k$ (where $k > 0$) is $\frac{32}{3}$. Find k .
- A** 8
B 3
C $\sqrt{8}$
D 4
E 2
F 16
- 9** The curve $y = x(x - 1)(x - 2)$ and the x -axis enclose two regions. What is the ratio of the area of the region for $0 \leq x \leq 1$ to the area for $1 \leq x \leq 2$?
- A** 1 : 2
B 2 : 1
C 1 : 3
D 1 : 4
E 1 : 1
F 3 : 1
- 10** f is a function defined for all real x and $\int_0^4 f(x) dx = 0$. Which of the following statements must be true?
- I** $f(x) = 0$ for some x with $0 \leq x \leq 4$.
II The area between the curve and the x -axis from 0 to 4 is 0.
III $\int_0^2 f(x) dx = -\int_2^4 f(x) dx$
- A** none of them
B I only
C II only
D III only
E I and II only
F I and III only
G II and III only
H I, II and III